

Aditya Prasad

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PROFESSIONAL SUMMARY

Fullstack engineer with production experience at OLA Electric building interactive data platforms (React, Arrow IPC), distributed backend systems (Django, Celery), and high-throughput data pipelines (Pandas, Parquet). Side projects span browser-based developer tooling (React + Monaco + Express), LLM-driven applications, and C++ systems internals.

EDUCATION

Indian Institute of Technology, Bombay

Bachelors of Technology in Chemical Engineering with a Minor in Computer Science

May 2025

8.1

EXPERIENCE

Full-Stack Software Engineer

June 2025 – April 2026

OLA Electric, Battery Intelligence Division

Bangalore, Karnataka

- Engineered a full-stack data exploration platform (**React + Python microservices + Arrow IPC**) for low-latency interactive display of **600k+** time-series datapoints and **50+** experiment params
- Rebuilt cell test orchestration from sequential scripts into a **Django + Celery** job-queue system with queue-per-test-type isolation, surfacing thousands of test workflows on a **DRF**-backed real-time coverage dashboard
- Designed a flexible manufacturing data model (**Django + DRF**) with generic process abstraction and JSON-based parameters (**100+**), enabling cross-process lineage tracking across **12** steps for heterogeneous process lines
- Built a high-volume time-series processing pipeline (**Pandas + multiprocessing + Parquet**) for **2000+** battery cells per batch with charge-discharge segmentation, cutting processing time from **20+** to **3** minutes

Advanced Application Engineering Intern

May 2024 – July 2024

Accenture

Bangalore, Karnataka

- Prototyped **GenAI** automations (LLM-based ticket summarization, query resolution, candidate screening drafts) for HR workflows within an **Oracle Fusion Cloud** environment serving a **30,000+** employee enterprise client
- Mapped HR processes across **20+** departments (payroll, hiring, absence management) and modeled automation impact to quantify manual effort reduction opportunities on a **\$5M** implementation engagement

PROJECTS

CSES Browser IDE | *React, Express, Monaco, TypeScript*

github.com/AdityaPrasad275/cses-ide

- Created an in-browser competitive programming **IDE**: **React + Monaco, Express**, and **g++** compilation
- Implemented a test-case runner with live **stdout/stderr** streaming and **sub-second** feedback on user submissions
- Authored **15+** tests with **Vitest** (frontend UI/API) and **Jest + Supertest** (backend compilation paths)

InteReview.ai | *React, Express, Node.js, LLM*

github.com/AdityaPrasad275/mock-interview-ai

- Developed a **2-agent** AI mock interview app: a reviewer LLM iteratively coaches answers before submission
- Crafted **2 role-separated** prompts: **LLM-summarized** resume, **10-message** sliding window, **JSON** outputs
- Engineered a **React + Express/Node.js** chat app with Figma UI and **3** LLM types (summarize, interview, review)

AtlasDB | *C++, B+ Tree*

github.com/AdityaPrasad275/AtlasDB

- Architected a database storage engine in **C++** with **4KB** pages, an **LRU** buffer pool, and slotted-page records
- Implemented a **B+ tree** index with node split/merge logic, ordered leaf traversal, and **O(logn)** key lookups
- Designed a **TableWithIndex** layer: **7,600x** faster point lookups and **200x** faster range scans on **100K** records

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

Backend & APIs: Django, Django REST Framework, Express, Node.js, FastAPI, Celery, REST APIs, Uvicorn

Frontend: React, Monaco Editor, HTML/CSS

Data & Storage: PostgreSQL, SQLite, Polars, Pandas, Parquet, Arrow IPC, multiprocessing

AI / LLM: Prompt engineering, structured outputs, tool-use, RAG, multi-agent workflows, Tavily

Systems & Tooling: Linux, Git, rclone, pdflatex; Vitest, Jest, Supertest, Playwright